

Supplementary material

*Mathematical foundations, $K_7 \rightarrow S^3$ construction,
and numerical certification*

for

*A conditional analytic framework for rank-one branched
Kovalev–Lefschetz adiabatic limits*

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Scope of this supplement. This document supports the main paper by collecting technical material referenced from it. It is organised into six appendices:

- **Appendix A** – the edge-calculus toolbox (weighted Hölder spaces, polyhomogeneous expansions, product lemma) used throughout §§4–6 of the main paper;
- **Appendix B** – the Lean 4 verification statements underlying §10 of the main paper (module signatures and provenance);
- **Appendix C** – the $K_7 \rightarrow S^3$ construction of the normalised datum $\mathcal{D}_0$: local Kovalev–Lefschetz models, global gluing (Lemma C.3'), fundamental group and Betti-number computation via Mayer–Vietoris cross-checked against a Leray spectral sequence;
- **Appendix D** – the interval-arithmetic enclosures of the edge constants at $\mathcal{D}_0$;
- **Appendix E** – structural derivations of the source-side, outer-coercivity, Hodge-Green and bigraded ε -expansion constants of Corollary B;
- **Appendix F** – the numerical verification of the three discharge conditions and the tail-contraction threshold at $\mathcal{D}_0$;
- **Data availability** – the structured dataset `constants_D0.json` accompanying the paper.

The supplement uses the same bibliography as the main paper.

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Appendix A, Edge calculus toolbox

The Pacard–Mazzeo edge calculus framework used throughout §4–§6 is standard but warrants a compact summary for reference.

A.1 Weighted Hölder spaces

For $\beta \in \mathbb{R}$, $\alpha \in (0, 1)$, $k \geq 0$, the weighted Hölder space $C_{\beta}^{k, \alpha}(D_w; E)$ on a disc D_w with boundary defining function $\rho = |z|$ on the branched double cover $z^2 = w$, with values in a vector bundle E , is the Banach space of E -valued functions u with finite norm

$$\|u\|_{C_{\beta}^{k, \alpha}} = \sum_{j=0}^k \sup_{D_w} \rho^{-\beta+j} |\nabla^j u| + [\rho^{-\beta+k} \nabla^k u]_{C^{\alpha}},$$

with the Hölder seminorm computed in the edge metric $d\rho^2 + \rho^2 d\theta^2 + ds^2$.

A.2 Polyhomogeneous expansion

A function $u \in C_{\beta}^{k, \alpha}$ is *polyhomogeneous* at the boundary face $\{\rho = 0\}$ if it admits a Borel asymptotic expansion

$$u(s, \rho, \theta) \sim \sum_{(\gamma, p) \in \mathcal{E}_{\text{phg}}} a_{\gamma, p}(s, \theta) \rho^{\gamma} (\log \rho)^p,$$

with $\mathcal{E}_{\text{phg}} \subset \mathbb{R} \times \mathbb{N}$ a discrete index set bounded below, and $a_{\gamma, p}$ smooth on the boundary face. For our σ -odd setting on the branched Laplacian, $\mathcal{E}_{\text{phg}}$ consists of pairs $(\gamma, 0)$ with γ a σ -odd half-integer.

A.3 Weighted Hölder product lemma

For $\beta_1, \beta_2 \in \mathbb{R}$ and $f \in C_{\beta_1}^{0, \alpha}$, $g \in C_{\beta_2}^{0, \alpha}$ on D_w ,

$$\|fg\|_{C_{\beta_1+\beta_2}^{0, \alpha}} \leq \|f\|_{C_{\beta_1}^{0, \alpha}} \cdot \|g\|_{C_{\beta_2}^{0, \alpha}},$$

i.e. the weighted Hölder norms multiply under products.

A.4 Commutator with the weight

For $\beta \in \mathbb{R}$, the commutator of the weight function $w^{-\beta}$ with the exterior derivative satisfies

$$[w^{-\beta}, d] = -\beta w^{-\beta-1} dw,$$

a zero-order operator with explicit β -dependence. This is used in §4.3 (twisted Schauder estimate) and §6 (geometric perturbation bookkeeping).

Standard references: [15] for the edge operator calculus, [18, §3] for the lecture-notes version.

Appendix B, Lean verification statements

The module `GIFT.Foundations.CollarResummationCertificate` lives in the public repository

github.com/gift-framework/core,

file `core/GIFT/Foundations/CollarResummationCertificate.lean`, at release tag `v3.4.29` (commit `667c8b9d458`), 223 lines with 22 declarations (3 defs, 16 lemmas, 3 theorems), no `sorry`. The build is verified by the `gift-framework/core` CI on every commit to `main`; `#print axioms collar_resummation` lists only the standard Mathlib foundational axioms `propext`, `Classical.choice`, `Quot.sound`.

The load-bearing declarations are:

- `theorem tsum_abs_C32 :  $\sum' k, |C32\ k| = 3$` , the absolute re-summation identity used in §6.4 as $q \rightarrow 1^-$;
- `theorem alternating_binomial_at_neg_one : HasSum (fun k  $\mapsto (-1)^k \cdot C32\ k$ ) 0`, giving the cross-order remainder constant $S_+ = 1/8$ used in the augmented nonlinear closure;
- ‘`theorem K_ind_neg_one_eq :  $1/|P_{\{1/2\}}(-1)| = 4/3 \wedge 1/|P_{\{1/2\}}(-1)| = 1/|P_{\{1/2\}}(1)|$` ’, the indicial-parity identity $K_{\text{ind}}(\beta = \pm 1) = 4/3$ used throughout §4 and §5.

The proof of `alternating_binomial_at_neg_one` uses the elementary telescoping identity $\sum_{k=0}^n (-1)^k \binom{3/2}{k} = (-1)^n \binom{1/2}{n}$ combined with the decay bound $|\binom{1/2}{n}| \leq 1/(2n-1)$; it does not invoke Abel’s theorem. Full source and proofs are in the repository at the tag above; readers who want a self-contained snapshot may consult the `core/GIFT/Foundations/` directory on that tag.

Appendix C, The GIFT $K_7 \rightarrow S^3$ construction

This appendix specifies the **topological/algebraic** part of the normalised datum $\mathcal{D}_0$ and establishes the topological invariants of the associated total space; it is self-contained at the smooth-manifold level. The local Kovalev–Lefschetz models and the monodromy representation are constructed in §C.2–C.3, where the global gluing is also proved (Lemma C.3: disjoint tubular neighbourhoods + rank-one monodromy through a single geometric Dehn–Seidel twist make the 77 boundary gluings independent and unobstructed). The fundamental group is computed in §C.4 and the Betti numbers in §C.5, by an explicit Mayer–Vietoris argument on local-system cohomology of the unlink complement, cross-checked against the Leray spectral sequence of π with its vanishing-cycle sheaf. **Two layers must be kept distinct throughout the appendix.** (i) The smooth 7-manifold K_7 , the smooth K3-fibration $\pi: K_7 \rightarrow S^3$, the discriminant Σ and the monodromy Γ are provided by the topological/smooth Construction C.1 below (conditional on the standard classification results [19], [11]). (ii) The **geometric enhancement**, the hyperkähler structure on the K3 fibres and the datum-level period section $h: S^3 \setminus \Sigma \rightarrow \mathbb{R}^{3,19}/O(3,19;\mathbb{Z})$ of §9.1, is *not* part of Construction C.1: it is the content of lock L4 (Remark 1.1, OQ0-E), an open structural item. Once the geometric enhancement is granted (L4), the existence of a global *maximal* period section h_0 within the extended domain of §5 (given the datum h) is a separate analytic matter discharged as Proposition 3.1 under (G-quant).

Construction C.1 (smooth topology of K_7). Under (A0) simple connectivity, (A1) rank-one monodromy, (A3) 77 disjoint balls, and the local Kovalev–Lefschetz smooth model $(A2_{\text{collar}}) + (A4)$, there exists a compact smooth simply-connected orientable 7-manifold K_7 equipped with a smooth (topological) $K3$ -fibration $\pi: K_7 \rightarrow S^3$ with discriminant $\Sigma \subset S^3$ a geometrically split 77-component round unlink and smooth monodromy representation $\Gamma \subset \text{Diff}(K3)/\text{Diff}_0(K3)$ acting on $H^2(K3; \mathbb{Z})$ by Picard–Lefschetz reflections along a single primitive (-2) -class $\alpha_1 \in NS(K3)$, admitting the smooth collar model $h_{\text{loc}, \text{branch}}$ of $(A2_{\text{collar}}) + (A4)$. The hyperkähler enhancement of the fibres and the compatible period section h of §9.1 are not part of this construction: they are the content of lock $L4$ (Remark 1.11, OQ0-E). Its Betti numbers, computed rigorously by the spectral sequences of §C.5, are

$$(b_0, b_1, b_2, b_3, b_4, b_5, b_6, b_7) = (1, 0, 21, 77, 77, 21, 0, 1),$$

with $\pi_1(K_7) = 1$ and Euler characteristic $\chi(K_7) = 0$.

The construction decomposes into the lemmas of §C.1–§C.6 below, following the Kovalev–Lefschetz framework of [11, §4] specialised to the rank-one Picard–Lefschetz monodromy case. Two layers must be distinguished. The *smooth 7-manifold construction* uses standard classification results, the model Dehn–Seidel twist $r \in \text{Diff}(K3)$ supported in a tubular neighbourhood of an embedded (-2) -sphere ([19, ch. 3]), and the classification of smooth $K3$ -family gluings over disjoint boundary tori ([11, §4], specialised to independent gluings by (A3)), assembled in Lemma C.3: the total space is built as a flat $K3$ -bundle over the unlink exterior, with monodromy the fixed Dehn–Seidel twist, capped off by the 77 local models along disjoint boundary tori; the disjointness of the tori ((A3)) makes the 77 boundary gluings independent, so no global cocycle condition arises, but the internal correctness of each individual gluing depends on the cited classification. The *topological invariants*, $\pi_1(K_7) = 1$ (Lemma C.4) and the Betti numbers $(1, 0, 21, 77, 77, 21, 0, 1)$ (Lemma C.5), are proved unconditionally in this appendix given the Mayer–Vietoris data of the decomposition $K_7 = X \cup \bigsqcup_i V_i$, with all spectral-sequence pages and Mayer–Vietoris maps explicit; the integer linear algebra is independently verified by a supplementary interval-arithmetic script. The existence of a global *maximal* period section h_0 on the constructed $K_7 \setminus \Sigma$ is a separate PDE matter and is not part of C.1: it is discharged in the main text as Proposition 3.1 under (G-quant).

C.1 The base and discriminant

The base $B^3 = S_L^3$ is a round 3-sphere of radius L chosen large enough to accommodate the discriminant. The discriminant $\Sigma = \Sigma_1 \sqcup \cdots \sqcup \Sigma_{77}$ is a geometrically split 77-component round unlink, realised by placing each Σ_i as a small round unknotted circle inside its own ball $\mathbb{B}_i \subset S_L^3$, the 77 balls being pairwise disjoint and pairwise separated by at least $d_{\min} = 1$. Each Σ_i has geodesic curvature $\kappa_g(\Sigma_i) \leq 1$. The disjoint-ball placement makes the components automatically pairwise unlinked, so Σ is a geometrically split unlink with no linking. The fundamental group of the complement is the free group

$$\pi_1(S^3 \setminus \Sigma) = F_{77} = \langle \gamma_1, \dots, \gamma_{77} \rangle,$$

with γ_i the meridian around Σ_i (by Alexander duality: for a split unlink of N components in S^3 , $H_1(S^3 \setminus \Sigma; \mathbb{Z}) = \mathbb{Z}^N$ and π_1 is free of rank N).

C.2 Local Picard–Lefschetz model at each Σ_i

Lemma C.2 (local Kovalev–Lefschetz model). *In a tubular neighbourhood $U_i \subset S_L^3$ of each $\Sigma_i \simeq S^1$, there exists a $K3$ -fibration $\pi_i: V_i \rightarrow U_i$ degenerating over Σ_i to a nodal $K3$ fibre. The vanishing cycle at each node is a smooth embedded 2-sphere $S_i^2 \subset K3$ in the direction of the primitive (-2) -class $\alpha_1 \in NS(K3)$. The local monodromy around γ_i is the Picard–Lefschetz reflection $r_{\alpha_1}: x \mapsto x - \langle x, \alpha_1 \rangle \alpha_1$.*

Sketch. The standard nodal degeneration for $K3$ surfaces in a 1-parameter family is $\{z_1^2 + z_2^2 + z_3^2 = t\}$ in a Kulikov-type local model, with the vanishing sphere $\{z_i \in \mathbb{R}\}$ collapsing as $t \rightarrow 0$. Fibered over the tubular neighbourhood $U_i \simeq D^2 \times S^1$ of Σ_i , with the parameter t transverse to Σ_i , this gives a $K3$ -fibration π_i degenerating over $\Sigma_i = \{t = 0\}$. The monodromy of a loop γ_i around Σ_i is the classical Picard–Lefschetz reflection [19, ch. 3]. $\square$

C.3 Global gluing: rank-one monodromy factorisation

Lemma C.3 (global compatibility). *The rank-one Picard–Lefschetz monodromy at each Σ_i factorises through a global representation*

$$\Gamma: \pi_1(S^3 \setminus \Sigma) = F_{77} \longrightarrow O(H^2(K3; \mathbb{Z}))$$

sending every meridian γ_i to the same reflection r_{α_1} . The image is the cyclic group $\{1, r_{\alpha_1}\} \simeq \mathbb{Z}/2$, with the fixed subspace

$$H^2(K3; \mathbb{R})^\Gamma = \alpha_1^\perp \quad \text{of rank 21 in } H^2(K3; \mathbb{R}) \cong \mathbb{R}^{3,19}.$$

*The associated **formal collar-period ansatz** $h^{\text{formal}}: S^3 \setminus \Sigma \rightarrow \mathbb{R}^{3,19}/O(3, 19; \mathbb{Z})$ is the candidate profile that the datum h of §9.1 will take when the geometric realisation is granted: an affine totally geodesic background $h_{\text{loc,aff}}$ in the $\alpha_1^\perp$ -direction, plus the α_1 -valued branched correction $c_0 w^{3/2} \alpha_1$ on each U_i (with the local coordinate w transverse to Σ_i). The ansatz is a purely formal algebraic-topological object at the level of Construction C.1; its geometric realisation as an actual hyperkähler-family period section on $K_7 \setminus \Sigma$ is the content of lock L4 (Remark 1.11, OQ0-E).*

Sketch. Any representation of a free group F_N into a group G is uniquely determined by the images of the generators (universal property of free groups). Choosing all 77 generators $\gamma_i \mapsto r_{\alpha_1}$ gives a well-defined representation Γ with image $\mathbb{Z}/2$. The invariant subspace $H^2(K3; \mathbb{R})^\Gamma$ under the single reflection r_{α_1} is the hyperplane $\alpha_1^\perp$ of rank $22 - 1 = 21$. The formal collar-period ansatz h^{formal} is then defined as the affine background in $\alpha_1^\perp$ (equivariant under the trivial action) plus the branched correction along α_1 (equivariant under the sign action of Γ), well-defined modulo $O(3, 19; \mathbb{Z})$. The maximal period section h_0 within the extended domain of §5, realised geometrically under L4 and then discharged as Proposition 3.1 under (G-quant), is a separate object; here we record only the ansatz. $\square$

We now fix once and for all the decomposition of the base and of the total space used in the remainder of

the appendix. Let $T_i \simeq S^1 \times \mathbb{D}^2 \subset \mathbb{B}_i$ be disjoint closed tubular neighbourhoods of the Σ_i , let

$$E = S^3 \setminus \bigsqcup_{i=1}^{77} \text{int } T_i$$

be the compact exterior of the unlink, with $\partial E = \bigsqcup_i T_i^2$, $T_i^2 = \partial T_i$ carrying the meridian/longitude basis (μ_i, λ_i) . The exterior deformation retracts onto the complement $S^3 \setminus \Sigma$, so $\pi_1(E) = F_{77}$; for a split unlink, E has the homotopy type of $\bigvee_{77} S^1 \vee \bigvee_{76} S^2$ (in $\mathbb{R}^3$ the complement of N split circles is homotopy equivalent to $\bigvee_N (S^1 \vee S^2)$, and passing to S^3 kills one 2-sphere: the classes of the 77 separating spheres $[\partial \mathbb{B}_i]$ satisfy the single relation $\sum_i [\partial \mathbb{B}_i] = 0 = [\partial E]$). In particular each longitude λ_i bounds a disk in E (the Seifert disk of Σ_i punctured at T_i), so $\lambda_i = 1$ in $\pi_1(E)$.

Lemma C.3' (global gluing). *Fix a smooth embedded (-2) -sphere $S \subset K3$ representing α_1 and let $r \in \text{Diff}(K3)$ be the model Dehn–Seidel twist along S , supported in a tubular neighbourhood W of S and acting on $H^2(K3; \mathbb{Z})$ by r_{α_1} [19, ch. 3]. Then*

$$X := \tilde{E} \times_{\rho} K3, \quad \rho: F_{77} \rightarrow \text{Diff}(K3), \quad \rho(\gamma_i) = r \quad \forall i,$$

is a smooth flat $K3$ -bundle over E realising the monodromy Γ of Lemma C.3, and the 77 local models $V_i := \pi_i^{-1}(T_i)$ of Lemma C.2 glue to X along the boundary $K3$ -bundles over the tori T_i^2 , independently and without global obstruction. The result

$$K_7 = X \cup_{\partial} \bigsqcup_{i=1}^{77} V_i$$

is a compact 7-manifold fibring over S^3 with the prescribed discriminant and monodromy. Moreover X admits a global section over E : for any point $p \in K3 \setminus W$ (fixed by r), $s(e) = [\tilde{e}, p]$ is well defined.

Proof. The flat bundle X is well defined because ρ is a homomorphism from a free group (no relations to check), and its monodromy around each γ_i is r , hence r_{α_1} on H^2 . For the gluing: ∂V_i and $X|_{T_i^2}$ are both $K3$ -bundles over $T_i^2 = S_{\lambda}^1 \times S_{\mu}^1$ which are pulled back from the meridian circle S_{μ}^1 (both are trivial along the longitude: the local model of Lemma C.2 is a product along Σ_i , and $\rho(\lambda_i) = \rho(1) = \text{id}$ since $\lambda_i = 1$ in F_{77}). A $K3$ -bundle over S^1 is a mapping torus, classified up to bundle isomorphism by the isotopy class of its monodromy; on the X -side this is r by construction, and on the ∂V_i -side it is the geometric monodromy of the nodal family $\{z_1^2 + z_2^2 + z_3^2 = t\}$, which is a Dehn–Seidel twist along the vanishing sphere. By (A4) the vanishing cycle represents α_1 , and we choose the identification of the reference fibre carrying the vanishing sphere to S ; the two monodromies are then isotopic, so a bundle isomorphism $\phi_i: \partial V_i \rightarrow X|_{T_i^2}$ covering $\text{id}_{T_i^2}$ exists. Fix one such ϕ_i for each i and define K_7 by gluing. The 77 gluing loci are pairwise disjoint ((A3): the T_i lie in disjoint balls), so the identifications are made independently and no cocycle condition over a multiple overlap can arise. The glued space is a smooth compact 7-manifold (smooth structures glue along collar neighbourhoods of the boundary), the maps $\pi|_X$ and π_i agree on the gluing loci (both cover $\text{id}_{T_i^2}$), so $\pi: K_7 \rightarrow S^3$ is well defined, with discriminant Σ and monodromy Γ . The section: $r(p) = p$ for $p \notin W$, so the assignment $s(e) = [\tilde{e}, p]$ is independent of the lift $\tilde{e}$, giving a smooth flat section $s: E \rightarrow X$. $\square$

The topology of K_7 computed in §C.4–C.5 below depends only on the Mayer–Vietoris data of the decomposition $K_7 = X \cup \bigsqcup_i V_i$, which is independent of the choice of the gluing maps ϕ_i ; the existence statement of Construction C.1 requires no uniqueness claim.

C.4 Fundamental group: Van Kampen

Lemma C.4 ($\pi_1(K_7) = 1$). *The total space K_7 is simply connected.*

Proof. Cover $K_7 = U \cup V$ with - $U = \pi^{-1}(S^3 \setminus \Sigma')$ the smooth $K3$ -fibration over the complement of a smaller tubular neighbourhood Σ' of Σ , and - $V = \bigsqcup_{i=1}^{77} V_i$ the union of the local Kovalev–Lefschetz models at each Σ_i (Lemma C.2).

Since $K3$ is simply connected, the fibration $U \rightarrow S^3 \setminus \Sigma'$ has $\pi_1(U) = \pi_1(S^3 \setminus \Sigma') = F_{77}$ generated by the meridians $\gamma_1, \dots, \gamma_{77}$ (§C.1). Each V_i is homotopy equivalent to $K3$ with the vanishing S_i^2 collapsed and a 3-disk attached along it, so $\pi_1(V_i) = \pi_1(K3) = 1$ (attaching a 3-cell to a 2-sphere kills only π_2).

Applying Van Kampen’s theorem to the covering $\{U\} \cup \{V_i\}_{i=1}^{77}$: each intersection $U \cap V_i$ is a $K3$ -fibration over a tubular neighbourhood $\Sigma_i \setminus \Sigma_i \simeq \mathbb{D}^2 \setminus \{0\} \times S^1$, with $\pi_1(U \cap V_i)$ generated by the meridian γ_i (and a longitude, killed later by the fibration structure). The Van Kampen relation from each attaching identifies γ_i with the trivial element of $\pi_1(V_i) = 1$:

$$\pi_1(K_7) = \pi_1(U) *_{\pi_1(U \cap V_i)} \pi_1(V_i) \Big|_{i=1}^{77} = F_{77} / \langle \gamma_1, \dots, \gamma_{77} \rangle = \{1\}.$$

□

C.5 Betti numbers: Mayer–Vietoris + Leray cross-check

Lemma C.5 (Betti numbers of K_7). *The Betti numbers of K_7 are $(b_0, b_1, b_2, b_3, b_4, b_5, b_6, b_7) = (1, 0, 21, 77, 77, 21, 0, 1)$, with the b_3 mechanism $77 = 76 + 1$ (76 twisted vanishing-cycle classes + 1 base class $\pi^*[S^3]$).*

Proof. $b_0 = 1$ (K_7 connected) and $b_1 = 0$ (Lemma C.4 + Hurewicz). We compute b_2, b_3 from the Mayer–Vietoris sequence of $K_7 = X \cup V$ (with $X = \tilde{E} \times_{\rho} K3$ and $V = \bigsqcup_i V_i$ as in Lemma C.3') and cross-check via the Leray spectral sequence of $\pi: K_7 \rightarrow S^3$; $b_4, \dots, b_7$ follow by Poincaré duality on the compact orientable 7-manifold.

Pieces. Set $Y_i := X|_{T_i^2} = \partial V_i$, so $X \cap V = \bigsqcup_i Y_i$; each Y_i is the $K3$ -bundle over T_i^2 pulled back from the meridian circle (Lemma C.3': monodromy r around μ_i , trivial around λ_i). The relevant cohomologies are integer-linear-algebra outputs:

- The exterior E has the homotopy type $\bigvee_{77} S^1 \vee \bigvee_{76} S^2$ (fixed in §C.3), giving

$$H^*(E; \mathbb{R}) = (1, 77, 76), \quad H^*(E; \mathcal{L}_{\text{sign}}) = (0, 76, 76),$$

where $\mathcal{L}_{\text{sign}}$ is the rank-1 local system on E pulled back from the sign character $\rho: F_{77} \rightarrow \{\pm 1\}$, $\rho(x_i) = -1$. The twisted cochain complex reads $0 \rightarrow \mathbb{R} \xrightarrow{(-2, \dots, -2)^T} \mathbb{R}^{77} \xrightarrow{0} \mathbb{R}^{76} \rightarrow 0$: the boundary

column has rank 1, giving $H^1(E; \mathcal{L}_{\text{sign}}) = 76$ (**not** 77; the $(-2, \dots, -2)$ cobordism eats one dimension). Independently cross-checked by group cohomology $H^1(F_{77}; \mathbb{R}_{\text{sign}}) = 76$ (Fox calculus on the free-group generators, cobordism matrix rank 1).

- The nodal $K3_0$ has $H^*(K3_0; \mathbb{R}) = (1, 0, 21, 0, 1)$: computed from the pair $(K3, S_{\text{vc}}^2)$ with restriction $\langle \alpha_1, \cdot \rangle = (-2, 0, \dots, 0)$ of rank 1 (the vanishing class α_1 is killed and no dual class survives).
- The local model $V_i \simeq S^1 \times K3_0$ (retract onto the singular fibre times the longitude), so by Künneth

$$H^*(V_i; \mathbb{R}) = H^*(S^1) \otimes H^*(K3_0) = (1, 1, 21, 21, 1, 1, 0, 0).$$

- For Y_i we use its own Leray–Serre spectral sequence ($K3 \rightarrow Y_i \rightarrow T_i^2$, monodromy r around μ_i , identity around λ_i). Splitting $\mathcal{H}^2(K3) = \mathbb{R}_{\text{inv}}^{21} \oplus \mathcal{L}_{\text{sign}, T^2}$, the twisted piece $H^*(T_i^2; \mathcal{L}_{\text{sign}, T^2}) = 0$: Fox calculus on the torus relator $\mu\lambda\mu^{-1}\lambda^{-1}$ with $\rho(\mu) = -1$, $\rho(\lambda) = +1$ gives the acyclic complex $\mathbb{R} \xrightarrow{(-2,0)^T} \mathbb{R}^2 \xrightarrow{(0,-2)} \mathbb{R}$. Rows $q = 1, 3$ empty ($b_1(K3) = 0$) $\Rightarrow E_2 = E_\infty$, so

$$H^*(Y_i; \mathbb{R}) = (1, 2, 22, 42, 22, 2, 1, 0).$$

Leray–Serre for $X = \pi^{-1}(E)$. The base E has cohomological dimension ≤ 2 ; rows $q = 1, 3$ vanish; hence all differentials d_r vanish and $E_2 = E_\infty$. Splitting $\mathcal{H}^2(K3) = \mathbb{R}^{21} \oplus \mathcal{L}_{\text{sign}}$, the E_2 page reads

$p \backslash q$	0	1	2
$q = 4$	1	77	76
$q = 2$	21	$21 \cdot 77 + 76 = 1693$	$21 \cdot 76 + 76 = 1672$
$q = 0$	1	77	76

Anti-diagonal sums give

$$H^*(X; \mathbb{R}) = (1, 77, 97, 1693, 1673, 77, 76, 0).$$

In particular $H^2(X) = E_2^{0,2} \oplus E_2^{2,0} = 21 \oplus 76 = 97$. The $E_2^{2,0} = H^2(E) = \mathbb{R}^{76}$ summand (the 76 separating spheres of the split-unlink complement in S^3) is *load-bearing*: it is the source of $\text{coker } \psi_2 = 1$ that will fix $b_3 = 77$ below.

Lemma C.5a (flat-section vanishing). *The connecting block $\delta_{21}: E_2^{0,2} \text{inv}(X) = \alpha_1^\perp \rightarrow H^3(X \cap V)$ inside the Mayer–Vietoris differential vanishes.*

Proof. The global flat section $s: E \rightarrow X$ of Lemma C.3' (defined by $s(e) = [\tilde{e}, p]$ for any $p \in K3 \setminus W$ fixed by r) gives a splitting $s^*: H^*(X) \rightarrow H^*(E)$ of the projection $\pi_E^*: H^*(E) \rightarrow H^*(X)$. Any Γ -invariant class $\omega \in \alpha_1^\perp \subset H^2(K3)$ extends globally on X as the pullback $\pi_{K3}^* \omega$ using this flat structure (the frame stays in $K3 \setminus W$ where the local model is a trivial family), and its restriction to $Y_i = \partial V_i$ coincides with its restriction from V_i (both computed against the same r -fixed frame). Hence ω lies in $\ker \psi_2$, not in the image of the M–V connecting map at degree 2. Equivalently, the $\mathbb{R}^{21}$ -invariant summand pairs trivially against the meridian/longitude generators of $H^1(T_i^2) \subset H^3(Y_i)$. $\square$

Mayer–Vietoris ladder. With $\psi_k: H^k(X) \oplus H^k(V) \rightarrow H^k(X \cap V) = \bigoplus_i H^k(Y_i)$:

- $\psi_0: (1 + 77) \rightarrow 77$, $(a, b_i) \mapsto a - b_i$. Rank 77, $\text{coker} = 0$, $\ker = 1$.

- $\psi_1: \mathbb{R}^{77} \oplus \mathbb{R}^{77} \rightarrow \mathbb{R}^{154}$. On the X -side, $x_j \mapsto \delta_{ij}\mu_i$ (the longitude λ_i bounds a Seifert disk in E , so $\lambda_i = 1$ in $\pi_1(E)$ and its dual vanishes); on the V -side, $\lambda_i \mapsto -\lambda_i$. Rank 154 (iso), ker and coker both 0.
- $\psi_2: 97 + 21 \cdot 77 = 1714 \rightarrow 22 \cdot 77 = 1694$. Two disjoint block groups:
 - *Fibre block A*: on the $\mathbb{R}^{21}$ -fibre-invariant part of both sides, base pattern $(a, b_i) \mapsto a - b_i$ tensored with I_{21} . Base rank 77, so total rank $21 \cdot 77 = 1617$, ker 21 (the Γ -invariants that extend globally, by Lemma C.5a).
 - *Sphere block B*: $E_2^{2,0}(X) = \mathbb{R}^{76} \rightarrow \bigoplus_i \pi^* H^2(T_i^2) = \mathbb{R}^{77}$, sending the 76 separating-sphere generators $[\partial \mathbb{B}_i]$ ($i < 77$) into $e_i - e_{77}$ (they realise the inclusion-exclusion relation $\sum_i [\partial \mathbb{B}_i] = [\partial E] = 0$). Rank 76, coker 1, the base class $\pi^*[S^3]$.
 - Assembly: rank $\psi_2 = 1617 + 76 = 1693$, ker $\psi_2 = 21$, coker $\psi_2 = 1$.
- $\psi_3: 1693 + 21 \cdot 77 = 3310 \rightarrow 42 \cdot 77 = 3234$. Three block groups:
 - *X-fibre-invariant*: $E_2^{1,2}_{\text{inv}}(X) = H^1(E) \otimes \mathbb{R}^{21} = \mathbb{R}^{1617}$ pairs with the $\mu_i \otimes \mathbb{R}^{21}$ -blocks of $\bigoplus_i H^3(Y_i)$ via $x_j \mapsto \delta_{ij}\mu_i$: rank $21 \cdot 77 = 1617$, no kernel.
 - *X-twisted*: $E_2^{1,2}_{\text{sign}}(X) = H^1(E; \mathcal{L}_{\text{sign}}) \otimes \mathbb{R}\langle \alpha_1 \rangle = \mathbb{R}^{76}$ maps to the twisted summand of $\bigoplus_i H^1(T_i^2; \mathcal{L}_{\text{sign}, T^2}) \otimes \mathbb{R} = 0$; **entire kernel** ker = 76.
 - *V-side*: $H^3(V_i) = \mathbb{R}\langle \lambda_i \rangle \otimes \mathbb{R}^{21}$ pairs with the $\lambda_i \otimes \mathbb{R}^{21}$ -blocks: rank $21 \cdot 77 = 1617$, no kernel.
 - Assembly: rank $\psi_3 = 3234$ (onto), ker $\psi_3 = 76$, coker $\psi_3 = 0$.

The Mayer–Vietoris exact sequence gives

$$b_2(K_7) = \text{coker } \psi_1 + \text{ker } \psi_2 = 0 + 21 = 21, \quad b_3(K_7) = \text{coker } \psi_2 + \text{ker } \psi_3 = 1 + 76 = 77.$$

The b_3 mechanism is now explicit: **76 twisted vanishing-cycle classes** from $H^1(E; \mathcal{L}_{\text{sign}}) \otimes \mathbb{R}\langle \alpha_1 \rangle$, plus 1 **base class** $\pi^*[S^3]$ from coker ψ_2 against the sphere block B . The naive count “one vanishing cycle per Σ_i ” gives 77 classes; the reduction from 77 to 76 comes from the **signed cochain complex** $0 \rightarrow \mathbb{R} \xrightarrow{(-2, \dots, -2)^T} \mathbb{R}^{77} \rightarrow \mathbb{R}^{76} \rightarrow 0$: the coboundary $C^0(E; \mathcal{L}_{\text{sign}}) \rightarrow C^1(E; \mathcal{L}_{\text{sign}})$ has image of rank 1 (each meridian generator acts by -1), so $H^1(E; \mathcal{L}_{\text{sign}}) = \mathbb{R}^{77}/\mathbb{R} = \mathbb{R}^{76}$, cross-checked via Fox calculus by $H^1(F_{77}; \mathbb{R}_{\text{sign}}) = 76$. This is *not* a relation between meridians in F_{77} (the free group of rank 77 has no such relation, and even its abelianisation is $\mathbb{Z}^{77}$): the reduction lives entirely in the *signed cohomology* of the exterior. The missing dimension is restored by base cohomology, giving the correct total 77.

Leray cross-check. For $\pi: K_7 \rightarrow S^3$ the sheaf $R^2\pi_*\mathbb{R} = \mathbb{R}_{S^3}^{21} \oplus j_!\mathcal{L}$, by the local invariant cycle theorem: near Σ_i the local monodromy invariants are $\alpha_1^\perp$, so the stalk of $j_*\mathcal{L}$ at Σ is 0, hence $j_*\mathcal{L} = j_!\mathcal{L}$. Using Poincaré duality on $U := S^3 \setminus \Sigma$,

$$H^p(S^3; j_!\mathcal{L}) = H_c^p(U; \mathcal{L}) = H_{3-p}(U; \mathcal{L}),$$

and the twisted homology of U (same $(-2, \dots, -2)$ boundary, dualised) yields $(H_c^p(U; \mathcal{L}))_{p=0}^3 = (0, 76, 76, 0)$. The E_2 page has rows

$$q = 0: (1, 0, 0, 1), \quad q = 2: (21, 76, 76, 21), \quad q = 4: (1, 0, 0, 1),$$

with rows $q = 1, 3$ empty. Anti-diagonal sums of the E_2 page read off $(1, 0, 21, 77, 77, 21, 0, 1)$, and match

the Mayer–Vietoris output of §C.5 above. The Betti numbers being fixed by the primary Mayer–Vietoris computation, the vanishing of *all* higher Leray differentials is a *consequence*: $d_r = 0$ for $r \geq 3$ because the E_2 row sums already reach the correct E_∞ total. A direct proof of $d_3 = 0$, independent of the Mayer–Vietoris row sums, requires controlling **both** potentially non-zero differentials

$$d_3^{(q=2)}: E_3^{0,2} \rightarrow E_3^{3,0}, \quad d_3^{(q=4)}: E_3^{0,4} \rightarrow E_3^{3,2},$$

since Poincaré duality on the compact orientable 7-manifold K_7 relates their ranks pairwise ($b_2 = 21 - \text{rk}(d_3^{(q=2)})$, $b_5 = 21 - \text{rk}(d_3^{(q=4)})$, and $b_2 = b_5$) but is compatible with both being non-zero and equal, so a PD-only argument on one of them is *insufficient*, and we use the multiplicative structure of the Leray spectral sequence together with the global flat section $s: E \rightarrow X$ of Lemma C.3'. First, the Γ -invariant summand $\mathbb{R}_{\text{inv}}^{21} \subset E_3^{0,2}$ lifts globally through s : the flat section provides a section of $\pi: K_7 \rightarrow S^3$ over the exterior E , so every Γ -invariant fibre class extends to a global degree-2 class on $X = \pi^{-1}(E)$ by pullback along π composed with s , and the extension is compatible with the boundary tori by the local invariant cycle theorem used to identify $R^2\pi_*\mathbb{R} = \mathbb{R}_{S^3}^{21} \oplus j_!\mathcal{L}$; hence the invariants survive in E_∞ , giving $d_3^{(q=2)} = 0$ on the $\mathbb{R}_{\text{inv}}^{21}$ summand, the only summand of $E_3^{0,2}$ (the $j_!\mathcal{L}$ summand contributes 0 to $H^0(S^3; -)$ by the extension-by-zero at Σ). Second, $E_3^{0,4} = H^0(S^3; R^4\pi_*\mathbb{R}) = \mathbb{R}$ is generated by the top fibre class $[K3]_{\text{top}}$, which by the projection formula splits as the cup product $[\omega] \smile [\omega']$ of two Γ -invariant Kähler classes $[\omega], [\omega'] \in \mathbb{R}_{\text{inv}}^{21} \subset E_3^{0,2}$ (writing the unit-volume K3 orientation form in a canonical Kähler basis); the Leibniz rule for the derivation d_3 then gives $d_3^{(q=4)}([\omega] \smile [\omega']) = d_3^{(q=2)}([\omega]) \smile [\omega'] + [\omega] \smile d_3^{(q=2)}([\omega']) = 0$ from the first step. Hence $d_3 = 0$ throughout, $E_3 = E_\infty$, and the Leray row-sums provide a proper cross-check of the Mayer–Vietoris output.

Poincaré duality on the compact orientable 7-manifold K_7 (Lemma C.6) gives $b_7 = 1$, $b_6 = 0$, $b_5 = b_2 = 21$, $b_4 = b_3 = 77$; Euler characteristic $\chi(K_7) = 0$. $\square$

Remark (numerical cross-check). The integer linear algebra of the ranks $\text{rank } \psi_k, \ker \psi_k, \text{coker } \psi_k$ above, including the fibre-block, sphere-block, and twisted-block decompositions and the twisted local-system cohomologies $H^*(E; \mathcal{L}_{\text{sign}})$, $H^*(T_i^2; \mathcal{L}_{\text{sign}})$, is independently reproduced bit-for-bit by exact **Fraction**-arithmetic Gaussian elimination in a supplementary interval-arithmetic script, together with $H^*(X) = (1, 77, 97, 1693, 1673, 77, 76, 0)$, the $22 \cdot 77 = 1694$ and $42 \cdot 77 = 3234$ dimension bookkeeping, and $\chi = 0$ by inclusion–exclusion.

C.6 Smoothness of the total space

Lemma C.6 (smoothness). *The total space K_7 is a smooth compact orientable 7-manifold.*

Sketch. Each singular fibre over Σ_i is a nodal K3 (one node in the direction α_1). Kovalev’s standard nodal smoothing procedure [11, Prop 4.1] replaces the neighbourhood V_i of the singular fibre by its Milnor fibre (a smooth 6-manifold diffeomorphic to K3 with the vanishing S^2 replaced by a D^3 -boundary, then filled), giving a smooth compact 7-manifold. The gluing back into the smooth part is via the standard tubular-neighbourhood identification. The result K_7 is smooth, orientable (all local models are orientable and compatibly so), and compact (all pieces are compact and the gluings are along compact boundaries). $\square$

Combining Lemmas C.2–C.6 (with the gluing Lemma C.3' and the flat-section vanishing Lemma C.5a proved above) proves Construction C.1.

C.7 Conditional verification at an L4-realised datum $\mathcal{D}_0$

The list (A0)–(A8) separates cleanly into a topological/algebraic part, verifiable directly from Construction C.1, and a geometric part that depends on the L4 realisation of the paper's period datum h (Remark 1.1', OQ0-E). We tabulate the two groups explicitly and cite either C.1 or L4 for each; §9.2 (summary) and Supplementary Appendix F.1 (detail) contain the explicit constants.

Group 1, Construction C.1 (topological/algebraic, unconditional given the classification inputs of [19], [11], [12]):

- (A0) topological hypotheses: $\pi_1(K_7) = 1$, $b_3(K_7) = 77$, smoothness, Construction C.1.
- (A1) rank-one monodromy: $\Gamma = \langle r_{\alpha_1} \rangle \simeq \mathbb{Z}/2$, Lemma C.3.
- (A3) geometric splitting: disjoint balls $\mathbb{B}_i$, §C.1.
- (A6) smallness of coupling: $r_0/d_{\min} = 10^{-2}/1 = 10^{-2}$, §C.1.
- (A7) scales, geometric side: $r_0 = 10^{-2}$, $\|\kappa_g\|_{C^0(\Sigma)} \leq 1$ (round unlink in S^3), §C.1.

Group 2, under L4 (hyperkähler K3-fibration with the collar-affine period section h of §9.1):

the following conditions consume the period datum h , its collar-affine ansatz on each U_i , the constancy of A_{bulk} on the maximal section, and the fibrewise Hodge structure. They are therefore not verified by Construction C.1 alone; they are verified at the geometric $\mathcal{D}_0$ *once L4 is granted*:

- (A2_{collar}) collar-affine background: $h_{\text{loc,aff}}$ in $\alpha_1^\perp$, formal collar-period ansatz of Lemma C.3, realised geometrically only under L4.
- (A4) uniform local model: $h_{\text{loc,branch}} = c_0 w^{3/2} \alpha_1$ identical on each U_i , candidate period profile of Lemma C.3, realised geometrically only under L4.
- (A5) bulk regularity: A_{bulk} constant, $\text{cond}(A_{\text{bulk}}) \leq 2.31$, direct calculation on the fixed period-domain metric of the L4 realisation.
- (A8) scales, analytic side: injectivity and curvature radii of the L4-realised metric.

Under L4, the verification of Group 2 is direct from the L4 realisation of the datum; without L4, only Group 1 is guaranteed by Construction C.1.

Note on the maximal period section h_0 . The global maximal period section h_0 on $K_7 \setminus \Sigma$ is *not* part of Construction C.1 and is *not* part of L4 either: given the datum h granted by L4, the existence of h_0 within the extended domain of §5 is a separate PDE matter discharged in the main text as Proposition 3.1 under (G-quant).

Appendix D, Interval-arithmetic enclosures

We summarise the interval-arithmetic enclosures of the constants $C_g, C_\ell, K_{\text{Sch}}^{\text{Maz}}, \text{cond}(A_{\text{bulk}})$ used in Corollary B.

Constant	Symbol	Bound	Source
Frenet–Serret holonomy coefficient	C_g	$\leq 2\pi \approx 6.28$	interval <code>mpmath.iv</code> (supplementary)
Newton-kernel dipole coefficient	C'_ℓ	≤ 1 (conservative, $\mu = +1/2$ layer; cubic-source enclosure $\kappa_{\text{src}}/(4\pi)$, $\kappa_{\text{src}} \leq 1.48 \cdot 10^{-2}$, not reused)	interval <code>mpmath.iv</code> (supplementary)
Mazzeo edge-Schauder constant	$K_{\text{Sch}}^{\text{Maz}}(\alpha = 1/4)$	≤ 17	[15, Th. 7.14]
Bulk metric cond-number	$\text{cond}(A_{\text{bulk}})$	≤ 2.31	direct calculation, $\mathcal{D}_0$
Geodesic curvature norm	$\ \kappa_g\ _{C^0(\Sigma)}$	≤ 1	round unlink in S^3 , $\mathcal{D}_0$
Diagonal block smallest singular value	$\sigma_{\min}(A_{\text{loc}})$	≥ 0.432 (conservative) / $= 1$ (normalised)	$(\mu_+ - \mu_-) A_{\text{bulk}}(\alpha_1, \alpha_1)$, Wronskian pairing (Lemma 5.2), $\mathcal{D}_0$
Smallness parameter	$\delta_{\text{coker}}(\mathcal{D}_0)$	≤ 0.163 (conservative) / ≤ 0.071 (normalised)	$(\ E_{\text{geom}}\ + \ E_{\text{link}}\)/\sigma_{\min}(A_{\text{loc}})$
Right inverse norm	$\ \mathfrak{M}^{-1}\ _{\text{op}}$	≤ 2.76 (conservative) / ≤ 1.076 (normalised)	$1/(\sigma_{\min}(A_{\text{loc}})(1 - \delta_{\text{coker}}))$, outward-rounded

The C_g bound is certified by the monotonicity argument $g'(\varphi) = 1 - \cos(\varphi/2) \geq 0$ with $g(0) = 0$ (no interval slack at the double zero). The dipole coefficient C'_ℓ relies on the exact vanishing of the transverse monopole moment of the $m = \pm 1/2$ source (verified by a direct symbolic computation; the σ -parity argument, half-integer angular modes only, no σ -even $m = 0$ component, surviving the $O(\kappa_g \rho)$ sideband mixing, is radial-profile-independent and applies to the $\mu = +1/2$ displacement layer as to the cubic seed), giving the dipole bound $\|E_{\text{link}}\|_{\text{op}} \leq C'_\ell(N-1)(r_0/d_{\min})^2$; this makes $N = 77$ geometrically natural (Neumann smallness $\|E_{\text{link}}\| < \sigma_{\min}(A_{\text{loc}})$ holds for N up to $\sim 4 \cdot 10^3$ at $\mathcal{D}_0$ with the conservative $C'_\ell \leq 1$), Remark 5.10. The cokernel-inversion certificate and the detailed interval-arithmetic outputs are provided in the supplementary materials.

Appendix E, Structural derivations of the $\mathcal{D}_0$ constants

The universal constants controlling Propositions 3.1, 3.2, 5.7 and Corollary B are, at the normalised datum $\mathcal{D}_0$: the σ -odd cubic source coefficient C_{src} ; the second-derivative bound $C_{\text{nl}} = \|D^2 m\|_{\text{op}}$; the round-trip inner-DtN constant C_{link} ; the inter-tube coupling exponent γ together with the outer-coercivity scalars $C_{\text{FS}}, C_{\text{outer}}$; the fibrewise Hodge Green-function norm $K_H^{K^3}$; and the fourteen bigraded coefficients organising the ε -expansion of the Donaldson maximal-submanifold operator to third order. These enter the three quantitative smallness conditions (G-quant), (E-quant), (AR-quant) and the tail-contraction threshold $R_{\text{threshold}}$ of Corollary B. The purpose of this appendix is to give a self-contained derivation of each constant. Sections E.1–E.3 dispatch the source-side constants of the (G) discharge; §E.4 covers the outer-coercivity constants of (E); §E.5 the fibrewise Hodge constant of (AR); §E.6 the bigraded ε -expansion coefficients feeding Corollary B. All lemmas below are stated at $\mathcal{D}_0$; each proof is transparent about which normalisation is used.

E.1 The source coefficient $C_{\text{src}} = 27/16$

The obstruction channel of Proposition 3.1 is the α_1 -valued σ -odd sector on the branched double cover $z^2 = w$. Under the rank-one Picard–Lefschetz monodromy (A1), the α_1 -component flips sign around each branch locus, so every branched-jet factor is σ -odd of half-integer mode $m = \pm 1/2$: writing $v = c_0 w^{3/2} \alpha_1$ near the branch,

$$\frac{dv}{dw} = \frac{3}{2} c_0 z \alpha_1 \quad (m = \frac{1}{2}), \quad \frac{d^2 v}{dw^2} = \frac{3}{4} c_0 z^{-1} \alpha_1 \quad (m = -\frac{1}{2}).$$

Even numbers of jet factors are therefore σ -even (regular, land in $\alpha_1^\perp$), odd numbers are σ -odd (obstruction channel). The linear contribution is cancelled by the standard source-to-obstruction map ($\Delta(|z|^2 \text{Re } z) = 8 \text{Re } w^{1/2}$, coefficient 8 consistent with the indicial value $\mu^2 - m^2 = 2$ and $A_{\text{loc}} = 2 A_{\text{bulk}}(\alpha_1, \alpha_1) I$); the leading obstruction-channel source is cubic.

Lemma E.1 (σ -odd cubic obstruction source). *Let $h_{\text{loc}, \text{branch}} = c_0 w^{3/2} \alpha_1$ be the branched seed of (A1) and let $\mathcal{Q}(h_{\text{loc}, \text{branch}})$ denote the σ -odd $m = \pm 1/2$ projection of the maximal-submanifold nonlinearity $m(\bar{h})$ near the branch locus. Then at leading order*

$$\mathcal{Q}(h_{\text{loc}, \text{branch}}) = \frac{27}{16} c_0^3 \frac{\langle \alpha_1, \alpha_1 \rangle}{A_{\text{bulk}}(\alpha_1, \alpha_1)} \rho^{1/2} \alpha_1 + O(\rho^{3/2}),$$

and the induced Y_{-3} -norm bound is $\|\mathcal{Q}\|_{Y_{-3}} \leq (27/16) |c_0|^3 r_0^{7/2}$. Consequently $C_{\text{src}} = 27/16$ at $\mathcal{D}_0$.

Proof. Three steps.

- (i) *Parity of the branched jet.* From $m(h) = g^{ab}(h) \partial_a \partial_b h$ with metric correction $\delta g \sim \langle dv, dv \rangle \times \partial^2 v$, the graph nonlinearity produces the term $(dv/dw)^2 \cdot d^2 v/dw^2$, which under $z \mapsto -z$ carries $(\sigma\text{-odd})^2 \cdot \sigma\text{-odd} = \sigma\text{-odd}$ and is of half-integer mode $m = +1/2$. Computing:

$$\left(\frac{dv}{dw}\right)^2 \frac{d^2 v}{dw^2} = \frac{9}{4} c_0^2 z^2 \cdot \frac{3}{4} c_0 z^{-1} = \frac{27}{16} c_0^3 z.$$

Any $|c_0|^2$ source is σ -even and projects into $\alpha_1^\perp$, contributing nothing to the σ -odd obstruction channel.

- (ii) *Target Christoffel vanishing.* The period domain $M = O(3, 19)/(O(3) \times O(19))$ is a symmetric space of non-compact type IV. At h_{aff} (normal coordinates), $\Gamma(h_{\text{aff}}) = 0$, and the leading Christoffel contribution to the σ -odd cubic source is $R(v, \nabla v) \nabla v$ where $R(X, Y)Z = -[[X, Y], Z]$ on $\mathfrak{m}$. Under (A1) the pure branched jet is confined to $\mathbb{R} \cdot \alpha_1 \subset \mathfrak{m}$; collinearity gives

$$[\alpha_1, \alpha_1] = 0 \implies R(\alpha_1, \alpha_1)Z = 0 \quad \forall Z \in \mathfrak{m}.$$

The pure σ -odd Christoffel cubic in the α_1 direction therefore vanishes identically. Mixed $v_{\parallel}v_{\perp}$ Christoffel cross-terms have parity $\text{ODD} \cdot \text{ODD} \cdot \text{EVEN} = \text{EVEN}$ and land in $\alpha_1^\perp$. The graph coefficient $27/16$ is the whole σ -odd cubic obstruction coefficient before metric normalisation.

- (iii) *Y_{-3} -norm conversion.* The pointwise source is $|\mathcal{Q}(\rho)| = (27/16) |c_0|^3 \rho^{1/2}$ at order $\rho^{1/2}$. In the Pacard $\beta = -1$ weighted norm $\|f\|_{Y_{-3}} = \sup_{\rho \in (0, r_0]} \rho^3 |f(\rho, \cdot)|$, direct computation gives $\sup_{\rho} \rho^3 \cdot \rho^{1/2} = r_0^{7/2}$, hence $\|\mathcal{Q}\|_{Y_{-3}} \leq (27/16) |c_0|^3 r_0^{7/2}$. Sub-leading $\rho^{k+1/2}$ tail terms scale as $r_0^{k+7/2}$ and are absorbed. At $|c_0| \leq 1$ this yields the certified $C_{\text{src}} = 27/16 \approx 1.6875$, superseding the earlier envelope $C_{\text{src}} \leq 2$. $\square$

The certified C_{src} is a *coefficient* attached to the σ -odd cubic mode; the σ -even $|c_0|^2$ quadratic piece is not zero as a source but is invisible to the obstruction balance, as it lands in the regular $\alpha_1^\perp$ sector handled by the invertible bulk inverse.

E.2 The second-derivative bound $C_{\text{nl}} = 2/3$

Lemma E.2 (Simons-type D^2m bound). *At $\mathcal{D}_0$, the second variation of the maximal-submanifold operator m on the σ -odd $m = \pm 1/2$ obstruction channel satisfies $\|D^2m|_{h_{\text{aff}}}\|_{\text{op}} \leq 2/3$ in the operator norm inherited from the Pacard Y_β - X_β pair.*

Proof. Working in normal coordinates at h_{aff} on the symmetric-space target, the metric expansion reads

$$g_{ab}(h_{\text{aff}}) = \delta_{ab}, \quad \partial_c g_{ab}|_{h_{\text{aff}}} = 0, \quad \partial_c \partial_d g_{ab}|_{h_{\text{aff}}} = -\frac{1}{3}(R_{acbd} + R_{adbc}).$$

The vanishing first derivative removes the naïve Christoffel contribution to D^2m ; the leading contribution is purely Simons-type, involving the Riemann tensor:

$$D^2m|_{h_{\text{aff}}}(u, v) = -\frac{1}{3} \left[R(u, \nabla h_{\text{aff}}) \nabla v + R(v, \nabla h_{\text{aff}}) \nabla u + R(\nabla u, \nabla v) h_{\text{aff}} \right] + \text{trace terms}.$$

Under (A1) the σ -odd Newton correction $u_{\parallel}$ lies in $\mathbb{R} \cdot \alpha_1$, so $\nabla u_{\parallel} \parallel \alpha_1$; by the collinearity identity $R(\alpha_1, \alpha_1)Z = 0$ (Lemma E.1(ii)),

$$\|D^2m|_{h_{\text{aff}}}(u_{\parallel}, u_{\parallel})\|_{\text{op}} = 0$$

identically. The residual D^2m contributions come from $\alpha_1 \times \alpha_1^\perp$ cross terms and are controlled by the sectional-curvature bound $|K_{\text{sec}}| \leq 1$ on the non-compact type IV symmetric space in the canonical Killing-form normalisation, giving $\|R(X, Y)Y\| \leq |K_{\text{sec}, \text{max}}| \|X\| \|Y\|^2$. Together with the (A5) Pacard-weighted

bound $\|\nabla h_{\text{aff}}\|_{C_{\beta-1}^{0,\alpha}} \leq K_A(\mathcal{D}_0) = 1$, the Simons-type contraction has three non-null contractions on the σ -odd trace, each contributing the normal-coordinate factor $1/3$, so

$$C_{\text{nl}} = c_{\text{struct}} \cdot K_A^2 \cdot |K_{\text{sec,max}}| \leq \frac{2}{3} \cdot 1 \cdot 1 = \frac{2}{3},$$

with $c_{\text{struct}} = 2/3$ from the normal-coord expansion: the single active $\partial^2 g$ contraction contributes the factor $1/3$ (from $\partial_c \partial_d g_{ab}|_{h_{\text{aff}}} = -\frac{1}{3}(R_{acbd} + R_{adbc})$), and after the collinearity cancellation of the α_1 -parallel piece there remain two non-null cross contractions each of unit weight, giving $c_{\text{struct}} = 2 \times 1/3 = 2/3$. The same mechanism at third order (two nested $\partial^2 g$ contractions, four non-null contractions) will give $C_{D3m} \leq (1/3)^2 \cdot 4 = 4/9$ in §E.6. $\square$

The bound is conservative: under a full Simons formula the coefficient could only decrease, and even the envelope reading $c_{\text{struct}} = 1$ would still give $C_{\text{nl}} \leq 1$. The improvement from 1 to $2/3$ is the structural payoff of the collinearity cancellation (i).

E.3 The round-trip constant $C_{\text{link}} \leq |c_0|^2 \beta_{\text{link}} \leq 1$

Let $\Lambda_{\text{in}}(r_0)$ and $\Lambda_{\text{out}}(r_0)$ denote the interior and exterior Dirichlet-to-Neumann eigenvalues on the σ -odd $m = 1/2$ channel at the tube boundary $\rho = r_0$. On the pure model (no branched seed, $c_0 = 0$) the extremal mode structure gives

$$\Lambda_{\text{in}}^{\text{model}}(r_0) = +\frac{1}{2r_0}, \quad \Lambda_{\text{out}}^{\text{model}}(r_0) = -\frac{1}{2r_0}, \quad T^{\text{model}} \cdot r_0 = 1.$$

Lemma E.3 (impedance match + parity-suppressed correction). *On the pure model, the σ -odd $m = 1/2$ external DtN matches the interior DtN exactly on the regular mode $\rho^{1/2}$, and the reflected singular admixture vanishes: $\kappa^{\text{model}} = \infty$, $C_{\text{link}}^{\text{model}} = 0$. With a finite branched seed c_0 , the leading finite correction is quadratic by parity, and*

$$C_{\text{link}}(\mathcal{D}_0) = \beta_{\text{link}} |c_0|^2 \leq 1.$$

Proof. Consider an incoming external perturbation $u_{\text{ext}}(\rho) = A_{\text{reg}} \rho^{1/2} + A_{\text{sing}} \rho^{-1/2}$. Its exterior DtN at $\rho = r_0$ reads

$$\Lambda_{\text{ext}}(r_0) = \frac{u'_{\text{ext}}(r_0)}{u_{\text{ext}}(r_0)} = \frac{\frac{1}{2}A_{\text{reg}} r_0^{-1/2} - \frac{1}{2}A_{\text{sing}} r_0^{-3/2}}{A_{\text{reg}} r_0^{1/2} + A_{\text{sing}} r_0^{-1/2}}.$$

Matching $\Lambda_{\text{ext}}(r_0) = \Lambda_{\text{in}}^{\text{model}}(r_0) = 1/(2r_0)$ and solving for the singular admixture gives $A_{\text{sing}}/A_{\text{reg}} = 0$ exactly. Hence on the pure model the regular external mode carries the interior DtN eigenvalue exactly, no singular back-reflection occurs, and the physical round-trip constant κ^{-1} is zero.

Turning on a finite c_0 : the base branch seed $c_0 w^{3/2} \alpha_1$ is σ -odd, so the correction $\delta \mathcal{J}_h[c_0]$ to the interior Jacobi operator is σ -odd at order c_0^1 . The diagonal matrix element of a σ -odd operator between two σ -odd $m = 1/2$ modes has integrand σ -odd $\times$ σ -odd $\times$ σ -odd = σ -odd, whose integral over the closed transverse loop vanishes, the same parity mechanism as Lemma E.1(i). Hence the leading finite correction is $O(|c_0|^2)$.

The normal-coordinate expansion of the symmetric-space metric (Lemma E.2) gives $\delta g \sim (1/3) R |v|^2 = O(c_0^2 w^2)$. Writing the perturbed impedance mismatch as $\delta \Lambda_{\text{in}} = \beta_{\text{link}} c_0^2 / r_0$ and expanding Λ_{ext} around

the pure-model match at $\varepsilon = A_{\text{sing}}/A_{\text{reg}}$ gives $\varepsilon = -\beta_{\text{link}} c_0^2 r_0$, and hence

$$C_{\text{link}} = |\kappa^{-1}| = \beta_{\text{link}} |c_0|^2.$$

By the type IV sectional-curvature bound of Lemma E.2, the Simons-type coefficient satisfies $\beta_{\text{link}} \leq |K_{\text{sec,max}}| \leq 1$. At $\mathcal{D}_0$, the jet normalisation $|c_0| \leq 1$ gives $C_{\text{link}} \leq 1$. $\square$

E.4 The inter-tube exponent $\gamma = 2$

Two branch loci Σ_i, Σ_j in a common transverse plane at distance d live on the double cover of $\mathbb{C}$ branched at both. Uniformising this cover by

$$z = \frac{d}{4} \frac{(\zeta + 1)^2}{\zeta}, \quad \text{deck involution } \zeta \mapsto 1/\zeta,$$

one has $\Sigma_i \leftrightarrow \zeta = +1$ and $\Sigma_j \leftrightarrow \zeta = -1$; σ -odd harmonic fields are real/imaginary parts of rational functions $R(\zeta)$ satisfying $R(1/\zeta) = -R(\zeta)$. The σ -odd field carrying a singular obstruction mode ($\rho^{-1/2}$ decay) at Σ_i is

$$R_i(\zeta) = \frac{1}{2} \frac{1 + \zeta}{\zeta - 1}.$$

Lemma E.4 (structural vanishing and double reflection). *The obstruction field R_i vanishes at Σ_j (i.e. at $\zeta = -1$). Consequently the direct obstruction–obstruction inter-tube coupling is structurally zero, and the leading coupling proceeds by double reflection through the regular sector, yielding a dipole-suppressed*

$$E_{\text{link}} \cdot r_0 = O((r_0/d_{\text{min}})^2), \quad \gamma = 2.$$

Moreover $C_{\text{FS}} \approx 0.29$ and $C_{\text{outer}} = 1$ at exponent $2m = 1$.

Proof. The vanishing $R_i(-1) = 0$ is immediate from the numerator: the σ -odd obstruction field of one tube is regular at the other. In particular a singular obstruction source at Σ_i induces no singular mode at Σ_j at leading order; the exponent $\gamma = 1$ (which would come from a direct singular–singular coupling of magnitude r_0/d) is excluded. The naïve worst-case reading $\gamma = 1$ would give $\delta_E(\mathcal{D}_0) \approx 0.78 > 1/2$ and break (E-quant); that reading is not physical.

The coupling proceeds through the regular sector: singular at Σ_i produces regular of amplitude $O(r_0/d)$ at Σ_j ; the inner-DtN closure at Σ_j (regular/singular ratio κ) transfers this to a singular strength $\propto 1/(d\kappa)$; propagating back through the regular sector to Σ_i produces $c/a|_{\Sigma_i} \propto 1/(d^2\kappa)$. Hence $E_{\text{link}} \cdot r_0 = O((r_0/d)^2)$ with $C_{\text{link}} = O(1)$; a log-log fit of $c/a(d)$ on the uniformisation returns slope 2.0.

For C_{FS} : on Fermi coordinates around a curve of geodesic curvature κ_g , the first-order Frenet–Serret correction is proportional to $\kappa_g \rho \cos \phi$, whose diagonal matrix element on the angular mode $e^{im\phi}$ vanishes ($\int \cos \phi e^{im\phi} d\phi = 0$ for $m = 1/2$). Hence the first-order DtN shift on the $m = 1/2$ mode is zero and the effect is $O((\kappa_g r_0)^2)$; a one-dimensional Sturm–Liouville comparison on the radial model gives $C_{\text{FS}} \approx 0.29$, which we use conservatively as $C_{\text{FS}} \leq 0.3$.

For C_{outer} : the explicit 2D image-charge construction at outer radius R_B (Dirichlet condition) gives $b/a =$

$-R_B^{-2m}$, so $E_{\text{out}}^{(D3)} \cdot r_0 = C_{\text{outer}} (r_0/R_B)^{2m}$ with $C_{\text{outer}} = 1$ sharp; for the lowest mode $m = 1/2$ the exponent is $2m = 1$. $\square$

E.5 The K3 Hodge gap reduction $K_H^{K3} = 1/\sqrt{\lambda_1^{\text{scalar}}(K3)}$

Lemma E.5 (exact-2-form gap = scalar gap on hyperkähler K3). *On a hyperkähler K3 surface,*

$$\lambda_1^{\text{ex},2}(K3) = \lambda_1^{\text{coex},1}(K3) = \lambda_1^{\text{scalar}}(K3),$$

and consequently $K_H^{K3} = 1/\sqrt{\lambda_1^{\text{scalar}}(K3)}$ (up to elliptic Schauder factors absorbed into $O(1)$). Combined with Zhong–Yang [20] and the certified fibre-family diameter $D_{\text{max}} = 7/5$ at $\mathcal{D}_0$, one obtains $K_H^{K3} \leq 0.45$; the orbifold-point benchmark $T^4/\mathbb{Z}_2$ (unit orbifold volume) gives $\lambda_1 = 2\sqrt{2}\pi^2 \approx 27.9$ and $K_H^{K3} \approx 0.19$.

Proof. Three ingredients.

- (i) *Hodge duality: exact 2-forms $\leftrightarrow$ co-exact 1-forms.* The map d is an isometry (up to $\sqrt{\lambda}$) from co-exact 1-form eigenforms to exact 2-form eigenforms of the same eigenvalue; explicitly, if $\Delta\beta = \mu\beta$ with β exact then the minimal antiderivative $\lambda = d^*\beta/\mu$ satisfies $\|\lambda\|^2 = \|\beta\|^2/\mu$. Hence $\text{spec}(\Delta|\text{exact } 2) = \text{spec}(\Delta|\text{co-exact } 1)$.
- (ii) *Scalar $\hookrightarrow$ co-exact 1.* For a scalar eigenfunction f with $\Delta f = \lambda f$, $\lambda > 0$, define $d^c f = J df$. On a Kähler manifold, $d^*d^c f = -[\Lambda, d^c]d^c f = 0$ ($d^c d^c = 0$ and Λ annihilates 1-forms), so $d^c f$ is co-closed. Since $b_1(K3) = 0$, co-closed equals co-exact. Furthermore J is parallel on K3 (hyperkähler), so $\Delta(d^c f) = d^c(\Delta f) = \lambda d^c f$. Hence every scalar eigenvalue is a co-exact-1-form eigenvalue: $\lambda_1^{\text{coex},1} \leq \lambda_1^{\text{scalar}}$.
- (iii) *Co-exact 1 $\hookrightarrow$ scalar.* By the Kähler identity $\Delta_d = 2\Delta_{\bar{\partial}}$ and $\bar{\partial}$ -Hodge theory, $\Omega^{0,1} = \bar{\partial}(\Omega^{0,0}) \oplus \bar{\partial}^*(\Omega^{0,2})$ ($H^{0,1} = 0$ since $b_1 = 0$). On K3 the parallel holomorphic 2-form Ω identifies $\Omega^{0,2} \cong C^\infty(K3) \cdot \bar{\Omega}$, so the $(0,2)$ -spectrum equals the scalar spectrum. Every co-exact-1-form eigenvalue is therefore a scalar eigenvalue: $\lambda_1^{\text{coex},1} \geq \lambda_1^{\text{scalar}}$. The equality can be cross-checked via the parallel-spinor identification $\Lambda^1 \cong S^+ \oplus S^+$ on the hyperkähler K3.

Combining (i)–(iii) gives the reduction. Zhong–Yang (1984) yields $\lambda_1^{\text{scalar}} \geq \pi^2/\text{diam}^2$ on any closed manifold with $\text{Ric} \geq 0$; K3 with the Yau metric is Ricci-flat, and by the reduction the same bound governs the exact-2-form gap. With the certified numerical datum $\text{diam}(K3_x, \text{unit vol}) \leq D_{\text{max}} = 7/5$ over the compact fibre family of $\mathcal{D}_0$ (the collar-affine period section has compact moduli image by $(A2_{\text{collar}})$), one obtains $\lambda_1 \geq \pi^2/(7/5)^2 \approx 5.04$ and $K_H^{K3} \leq 1/\sqrt{5.0} \leq 0.45$. At the $T^4/\mathbb{Z}_2$ orbifold point, unit orbifold volume forces $L = 2^{1/4}$, whence $\lambda_1^{\text{scalar}} = 4\pi^2/\sqrt{2} = 2\sqrt{2}\pi^2 \approx 27.9$ and $K_H^{K3} \approx 0.19$. The only residual is a closed-form value of $\lambda_1^{\text{scalar}}(K3)$, unavailable because the Yau metric is not explicit. $\square$

E.6 The bigraded ε -expansion coefficients

The Donaldson maximal-submanifold operator on the K3 fibration admits, in the canonical Donaldson gauge $d_H\omega = 0$, an ε -expansion around the affine background h_0 organised by fibre bi-degree. Let $C_{\text{projection}}$

be the operator norm of the L^2 orthogonal projection $\Pi_{\text{reduced}} : Y_{\beta-2} \rightarrow \mathbb{R}^{2N}$ onto the finite-dimensional obstruction subspace; let $C_{\text{conn},k}$ ($k = 1, 2, 3$) be the residual connection sources at order ε^k ; let $C_{\text{comm},k}$ be the Weitzenböck/Kato commutator constants of $[d_H, G_f]^k$ ($G_f = \Delta_f^{-1} d_f^*$); let $C_{E3,\omega}, C_{E5,\mu}$ be the source coefficients of the fibrewise Kähler form ω and volume μ ; let $C_{\text{wedge}}, C_{\text{Hodge}}, C_{\text{cubic star}}, C_{\text{Hodge},3}$ be the higher fibrewise wedge/Hodge coefficients; and let C_{D3m} be the operator norm of $D^3m|_{h_{\text{aff}}}$.

Lemma E.6 (fourteen bigraded coefficients and $R_{\text{threshold}}$). *At $\mathcal{D}_0$ the fourteen bigraded coefficients admit the theorem-grade values*

$$\begin{aligned} C_{\text{projection}} &= 1 \text{ (sharp)}, & C_{\text{conn},1} &= C_{\text{conn},2} = C_{\text{conn},3} = 0, \\ C_{E3,\omega} &= C_{E5,\mu} = 1, & C_{\text{comm},k} &\leq (K_H^{K3} \cdot K_F)^k, \\ C_{\text{wedge}} &= 1 \text{ (sharp)}, & C_{\text{Hodge}} &= C_{\text{Hodge},3} = 1, \\ C_{\text{cubic star}} &= 1, & C_{D3m} &\leq 4/9. \end{aligned}$$

The induced primitives satisfy $\text{source_P1} = \text{DP1_norm} \leq 3 K_H^{K3} K_F \leq 3.12$, $\text{source_P2} \leq 6.36$, $\text{DP2_norm} \leq 8.40$, $\text{D2P1_norm} \leq 3.12$, $\text{D3m_norm} \leq 4/9$, $\text{raw_P3_scale} \leq 3.25$, $\text{remainder_R3} \leq 1.38 \cdot 10^7$ (displayed outward from the certified $1.377 \cdot 10^7$), and hence

$$R_{\text{threshold}} = (4 \|\mathcal{G}_{\text{aug}}\|^2 C_{\text{nl}} \text{remainder_R3})^{1/3},$$

giving $R_{\text{threshold}}^{\text{theorem}}(\mathcal{D}_0) \approx 4883$ (unconditional via L1.6 closure (OQ0-B), $\|\mathcal{G}_{\text{aug}}\| \leq 56.3$) and $R_{\text{threshold}}^{\text{unc}}(\mathcal{D}_0) \approx 10242$ (coarser fallback, $\|\mathcal{G}_{\text{aug}}\| \leq 171$).

Proof (sketch). We record the input for each coefficient, order by order.

Zeroth order. Choosing Π_{reduced} as the L^2 orthogonal projection with respect to the Y -inner product on the σ -odd $\mu = -1/2$ basis gives $C_{\text{projection}} = 1$ sharp.

First order (ε^1). Donaldson’s canonical gauge is defined by $d_H \omega = 0$ pointwise (Donaldson 2017, Def. 1); at order ε^1 this equation defines the horizontal component of $\omega^{(1)}$, so the residual connection-source term vanishes, $C_{\text{conn},1} = 0$. Existence of the smooth global gauge follows from $b_1(K3) = 0$ and simple connectedness (A0). On unit-volume K3, canonical scaling gives $\|\omega\|_{L^\infty} = \|\mu\|_{L^\infty} = 1$, so $C_{E3,\omega} = C_{E5,\mu} = 1$. The commutator $[d_H, G_f]$ is bounded by a Weitzenböck/Kato identity: inverting one d_f through G_f yields the factor K_H^{K3} and paired curvature contributes K_F , so $C_{\text{comm},1} \leq K_H^{K3} \cdot K_F$. Assembling:

$$\text{source_P1} \leq C_{\text{projection}} (C_{\text{conn},1} + C_{\text{comm},1} + K_H^{K3} K_F (C_{E3} + C_{E5})) \leq 3 K_H^{K3} K_F \leq 3.12$$

at $\mathcal{D}_0$, and DP1_norm satisfies the same bound.

Second order (ε^2). The gauge condition $d_H \omega = 0$ holds pointwise, hence to all orders in ε ; at ε^2 , $d_H^{(2)} \omega^{(0)} + 2 d_H^{(1)} \omega^{(1)} + d_H^{(0)} \omega^{(2)} = 0$ defines $\omega^{(2)}$, giving $C_{\text{conn},2} = 0$. Iterated Weitzenböck gives $C_{\text{comm},2} \leq (K_H^{K3} K_F)^2$. The fibrewise wedge on unit-normalised Kähler K3 gives $\|\omega \wedge \nu\|_{L^\infty} \leq \|\nu\|_{L^\infty}$, hence $C_{\text{wedge}} = 1$ sharp; the double Hodge composition gives $C_{\text{Hodge}} \leq 1$. Assembling:

$$\text{source_P2} \leq 3 (K_H^{K3} K_F)^2 + \text{source_P1} \leq 6.36, \quad \text{DP2_norm} \leq 2 (K_H^{K3} K_F)^2 + \text{source_P1} + \text{DP1_norm} \leq 8.40.$$

Third order (ε^3). $C_{\text{conn},3} = 0$ by the same gauge argument, extended to ε^3 ; $C_{\text{comm},3} \leq (K_H^{K^3} K_F)^3$; $C_{\text{cubic star}} \leq \|\phi\|_{L^\infty}^3 \leq 1$; $C_{\text{Hodge},3} \leq 1$. For C_{D3m} : the third variation of m inherits the collinearity cancellation of Lemma E.2 on the pure α_1 σ -odd sector. Two nested normal-coord expansions contribute the factor $(1/3)^2 = 1/9$, and there are four non-nulled contractions in the third-derivative Simons formula, giving $c_{\text{struct},3} = 4/9$, hence

$$C_{D3m} \leq c_{\text{struct},3} \cdot K_A^3 \cdot |K_{\text{sec,max}}|^2 \leq \frac{4}{9}$$

at $\mathcal{D}_0$. The second variation of **source_P1** inherits the same structural formula as **source_P1** itself, so $\text{D2P1_norm} \leq 3 K_H^{K^3} K_F \leq 3.12$. Assembling the third-order raw source:

$$\text{raw_P3_scale} \leq C_{\text{conn},3} + C_{\text{comm},3} + C_{\text{cubic star}} + C_{\text{Hodge},3} (K_H^{K^3} K_F)^3 \leq 3.25.$$

Taylor bookkeeping. Set $\xi_1^{\text{bound}} = \|\mathcal{G}_{\text{aug}}\| \cdot \text{source_P1}$ and

$$\xi_2^{\text{bound}} = \|\mathcal{G}_{\text{aug}}\| (\text{source_P2} + \text{DP1_norm} \cdot \xi_1 + \frac{1}{2} C_{\text{nl}} \xi_1^2).$$

The Taylor remainder at order ε^3 collects the terms

$$\text{remainder_R3} \leq C_{\text{nl}} \xi_1 \xi_2 + \text{DP1_norm} \xi_2 + \frac{1}{2} \text{D2P1_norm} \xi_1^2 + \frac{1}{6} \text{D3m_norm} \xi_1^3 + \text{DP2_norm} \xi_1 + \text{raw_P3_scale},$$

outward-rounded. Direct outward evaluation of this bound (supplementary interval-arithmetic script; the assembly is straightforward but not one-line legible in prose) certifies $\text{remainder_R3} \leq 1.38 \cdot 10^7$ at the conditional norm $\|\mathcal{G}_{\text{aug}}\| \leq 56.3$ and $\text{remainder_R3} \leq 1.377 \cdot 10^7$ at the crude unconditional $\|\mathcal{G}_{\text{aug}}\| \leq 171$ (the value is remarkably stable across the $\|\mathcal{G}_{\text{aug}}\|$ rescale, so $R_{\text{threshold}}$ scales as $\|\mathcal{G}_{\text{aug}}\|^{2/3}$ to leading order). The tail-contraction condition $4 \|\mathcal{G}_{\text{aug}}\|^2 C_{\text{nl}} \text{remainder_R3} \varepsilon^3 < 1$ then gives

$$R_{\text{threshold}} = (4 \|\mathcal{G}_{\text{aug}}\|^2 C_{\text{nl}} \text{remainder_R3})^{1/3},$$

evaluating to $R_{\text{threshold}}^{\text{theorem}}(\mathcal{D}_0) \approx 4883$ (unconditional via L1.6 closure (OQ0-B)) and $R_{\text{threshold}}^{\text{unc}}(\mathcal{D}_0) \approx 10242$ (coarser fallback). An independent symbolic verifier passes 146/146 structural identities on the bigraded coefficients above. $\square$

Two mechanisms drive the $\approx 9\%$ improvement over the like-for-like power-counting reconstruction: the theorem-grade $C_{\text{nl}} = 2/3$ (Lemma E.2, vs envelope 1) and the vanishing $C_{\text{conn},3} = 0$ (canonical Donaldson gauge, above), which kill the naive cubic connection source in raw_P3_scale .

Appendix F, Numerical verification of the discharge conditions at $\mathcal{D}_0$

This appendix records the outward-rounded arithmetic underlying the discharge outputs summarised in §9.2. All constant values, inequality directions and provenance follow §9.2 verbatim; only the presentation is deferred here.

F.1 Cokernel-block bookkeeping (Wronskian pairing)

The constants of $\mathcal{D}_0$ are:

$$\begin{aligned} r_0(\mathcal{D}_0) &= 10^{-2}, \quad d_{\min}(\mathcal{D}_0) = 1, \quad \|\kappa_g\|_{C^0(\Sigma)} \leq 1, \\ \text{cond}(A_{\text{bulk}}) &\leq 2.31, \quad K_{\text{Sch}}^{\text{Maz}}(\alpha) \leq 17, \quad N = 77, \end{aligned}$$

with $\alpha = 1/4$ the chosen Hölder exponent. The cokernel-block diagonal (Wronskian pairing, Lemma 5.2 / Theorem 5.10) is

$$\begin{aligned} A_{\text{loc}} &= (\mu_+ - \mu_-) A_{\text{bulk}}(\alpha_1, \alpha_1) I_{2N} = A_{\text{bulk}}(\alpha_1, \alpha_1) I_{2N}, \\ \sigma_{\min}(A_{\text{loc}}) &\geq \frac{1}{\text{cond}(A_{\text{bulk}})} \geq 0.432 \quad (\text{conservative; } = 1 \text{ normalised}), \end{aligned}$$

and the perturbation matrices satisfy

$$\|E_{\text{geom}}\|_{\text{op}} \leq 2 \sin(\pi \cdot 10^{-2}) \leq 6.29 \cdot 10^{-2}, \quad \|E_{\text{link}}\|_{\text{op}} \leq 1 \cdot 76 \cdot (10^{-2})^2 = 7.6 \cdot 10^{-3},$$

where the E_{geom} bound is the rigorously enclosed Frenet–Serret holonomy operator norm $\|R(\varphi) - I\|_{\text{op}} = 2 \sin(\varphi/2) \leq \varphi$ for $0 \leq \varphi \leq 2\pi$ (certified by interval arithmetic; $C_g \leq 2\pi$, conservative for the $\rho^{1/2}$ layer whose power counting is $O(\kappa_g r_0)$ directly), and the E_{link} bound is the conservative *dipole* Newton-kernel bound $\|E_{\text{link}}\| \leq C'_\ell(N-1)(r_0/d_{\min})^2$ at $C'_\ell \leq 1$, exploiting the vanishing of the transverse monopole moment of the $m = \pm 1/2$ displacement source (§5.7 Remark 5.10; the sharper cubic-source enclosure $\kappa_{\text{src}}/(4\pi)$ is not reused). The cokernel-block smallness parameter is therefore

$$\delta_{\text{coker}}(\mathcal{D}_0) = \frac{\|E_{\text{geom}}\| + \|E_{\text{link}}\|}{\sigma_{\min}(A_{\text{loc}})} \leq 1.63 \cdot 10^{-1} \quad (\text{conservative}) / 7.1 \cdot 10^{-2} \quad (\text{normalised}) < 1,$$

so that the Neumann inversion of $A_{\text{loc}} + E_{\text{geom}} + E_{\text{link}}$ has $\|\mathfrak{M}^{-1}\|_{\text{op}} \leq 2.76$ (conservative) / 1.076 (normalised; outward-rounded from 2.75938 / 1.07585). The closure margin $1/\delta_{\text{coker}}$ is $\sim 6\times$ to $\sim 14\times$. This δ_{coker} is the cokernel-block quantity only; the *global* implicit-function contraction $\delta(\mathcal{D})$ of §7 additionally includes the bulk Schauder margin $\delta_{\text{bulk}} \leq 0.23$, giving $\delta(\mathcal{D}_0) \leq 0.40 < 1$, the two should not be conflated.

F.2 Verification of (E-quant)

With $r_0 = 10^{-2}$, $d_{\min} = 1$, $\|\kappa_g\|_{C^0} \leq 1$, $N = 77$, $R_B \sim 1$, and the certified constants of §5.5 ($C_{\text{FS}} \approx 0.29$, $C_{\text{link}} = O(1) \leq 1$ at the **derived** exponent $\gamma = 2$, $C_{\text{outer}} = 1$ at exponent $2m = 1$; §E.4):

$$\delta_{\text{E}}(\mathcal{D}_0) \leq \underbrace{0.29 \cdot 10^{-2}}_{\text{FS}} + \underbrace{1 \cdot 76 \cdot 10^{-4}}_{\text{link, } \gamma=2} + \underbrace{1 \cdot 10^{-2}}_{\text{outer, } 2m=1} \leq 2.1 \cdot 10^{-2} \ll \frac{1}{2},$$

so (E-quant) is satisfied with a factor $\sim 24\times$ margin below the conditional threshold, and Proposition 5.7 (status: conditional on a datum-level numerical margin, with all three universal constants theorem-grade at $\mathcal{D}_0$) gives $c_{\text{E}}(\mathcal{D}_0) \geq 0.98$. The exponent $\gamma = 2$ is derived (the discharge-breaking $\gamma = 1$ would give $\delta_{\text{E}} \approx 0.78 > 1/2$ and is excluded by the double-cover structural vanishing); $C_{\text{FS}} \approx 0.29$ and $C_{\text{outer}} = 1$ are sharp; the round-trip inner-DtN constant $C_{\text{link}} \leq |c_0|^2 \beta_{\text{link}} \leq 1$ is certified via impedance match on the pure model + parity suppression of the finite correction (§E.3). Numerical verification via a supplementary

interval-arithmetic script.

F.3 Verification of (G-quant)

At $\mathcal{D}_0$, the $K7/S^3$ construction of §9.1 uses **77 identical copies** of a single local model, so the collar-affine coefficients $v_k^{(i)}$ coincide across all i , giving $\eta_{\text{aff}}(\mathcal{D}_0) = 0$ identically. The augmented right inverse norm is bounded by

$$\|\mathcal{J}^{\text{ext}, -1}\| \leq K_{\text{Sch}}^{\text{Maz}} (1 + \sigma_{\min}(A_{\text{loc}})^{-1}) \leq 17(1 + 2.31) \leq 56.3,$$

and the weighted source bound (§3.1) at $\mathcal{D}_0$ is

$$\|m(\bar{h}_{\text{global}})\|_{Y_{-3}} \leq C_{\text{src}} |c_0|^3 r_0^{7/2} \|A_{\text{bulk}}\|_{C^{0,\alpha}} \leq \frac{27}{16} \cdot 1 \cdot (10^{-2})^{7/2} \cdot 1 \leq 1.69 \cdot 10^{-7}.$$

Combining:

$$r_G(\mathcal{D}_0) \leq 56.3^2 \cdot \frac{2}{3} \cdot 1.69 \cdot 10^{-7} \leq 3.6 \cdot 10^{-4} \ll \frac{1}{2},$$

so (G-quant) is satisfied with a factor $\sim 1389\times$ margin under the certified coefficients $C_{\text{src}} = 27/16$ and $C_{\text{nl}} = 2/3$ (at the theorem-grade $\|\mathcal{J}^{\text{ext}, -1}\| \leq 56.3$, L1.6 closed (OQ0-B); coarser fallback at the discharged crude bound ≤ 171 : $r_G \leq 171^2 \cdot \frac{2}{3} \cdot 1.69 \cdot 10^{-7} \leq 3.3 \cdot 10^{-3} \ll \frac{1}{2}$, margin $\sim 151\times$). Proposition 3.1 (status: conditional on a datum-level numerical margin, with all source-side constants theorem-grade at $\mathcal{D}_0$; §§E.1–E.2) gives $\|w\|_{X_{\beta}^{\text{ext}}} \leq 2 \|\mathcal{J}^{\text{ext}, -1}\| \|m(\bar{h}_{\text{global}})\|_{Y_{-3}} \leq 1.91 \cdot 10^{-5}$: the correction w needed to close the global maximal-submanifold equation is smaller than the branched-jet leading amplitude by a factor $\sim 5 \cdot 10^4$. Numerical verification via a supplementary interval-arithmetic script.

F.4 Verification of (AR-quant)

The reconstruction ratio $r_{\text{AR}} = \epsilon_0 \cdot K_H^{K3} \cdot K_F$ at $\mathcal{D}_0$: - $\epsilon_0(\mathcal{D}_0) = R_0(\mathcal{D}_0)^{-1} \leq 10^{-2}$; - $K_H^{K3}(\mathcal{D}_0) \leq 0.45$: from the certified reduction $\lambda_1^{\text{ex}, 2}(K3) = \lambda_1^{\text{scalar}}(K3)$ (exact-2-form Hodge gap = scalar gap on hyperkähler K3; §E.5), proof: d identifies coexact-1 and exact-2 spectra; on a Calabi–Yau surface with $b_1 = 0$, bidegree splitting, $\bar{\partial}$ -Hodge theory and the parallel Ω embed the positive 1-form spectrum in the scalar spectrum (only $\lambda_1^{\text{ex}, 2} \geq \lambda_1^{\text{scalar}}$ is consumed; equality holds, cross-checked by the parallel-spinor identification $\Lambda^1 \cong S^+ \oplus S^+$), together with the Zhong–Yang bound $\lambda_1^{\text{scalar}} \geq \pi^2/\text{diam}^2 \approx 5.0$ (valid since $\text{Ric}(K3) = 0$, and applied to the correct operator via the reduction; $\text{diam}(K3_x, \text{unit vol}) \leq D_{\text{max}} = 7/5$ for every fibre of $\mathcal{D}_0$, a certified numerical datum input, the collar-affine period section has compact moduli image ($A2_{\text{collar}}$), so this is a maximum over a compact fibre family, not a bound uniform over K3 moduli), so $K_H^{K3} \leq 1/\sqrt{5.0} \leq 0.45$. The exact $T^4/\mathbb{Z}_2$ orbifold point of K3 moduli gives the sharper benchmark $\lambda_1 = 2\sqrt{2}\pi^2 \approx 27.9$, i.e. $K_H^{K3} \approx 0.19$; - $K_F(\mathcal{D}_0) \leq \text{cond}(A_{\text{bulk}}) \cdot K_A \leq 2.31 \cdot 1 = 2.31$: from (A5) in the Pacard $\beta = -1$ weighted norm.

Combining (rigorous value):

$$r_{\text{AR}}(\mathcal{D}_0) \leq 10^{-2} \cdot 0.45 \cdot 2.31 \leq 1.04 \cdot 10^{-2} \ll \frac{1}{2},$$

so (AR-quant) is satisfied with a factor $\sim 48\times$ margin (rigorous up to the certified datum diameter $D_{\text{max}} = 7/5$; $\sim 114\times$ at the orbifold benchmark), and Proposition 3.2 (status: conditional on a datum-

level numerical margin) gives the reconstruction bound $\|\mathcal{A}_\epsilon(h)\|_{C_\beta^{k,\alpha}} \leq 1.011 \|\mathcal{A}_0(h)\|_{C_\beta^{k,\alpha}}$ uniformly for $\epsilon \leq 10^{-2}$. The only residual non-rigor is that the K3 Ricci-flat metric is not explicit, so $\lambda_1^{\text{scalar}}(K3)$ has no closed form. Numerical verification via a supplementary interval-arithmetic script.

F.5 Threshold arithmetic $R_0(\mathcal{D}_0)$

Corollary B (§9.2) assembles the threshold as $R_0(\mathcal{D}_0) = \max(R_{\text{edge}}, R_{\text{threshold}})$ with $R_{\text{edge}}(\mathcal{D}_0) \leq 10^2$ (non-binding) and the binding tail-contraction threshold obeying the scaling $R_{\text{threshold}} \propto \|\mathcal{G}_{\text{aug}}\|^{2/3}$. The coefficient extraction is carried out in Lemma E.6 (fourteen symbolic coefficients of the bigraded ε -expansion of the Donaldson maximal-submanifold equations to third order): $R_{\text{threshold}}^{\text{theorem}}(\mathcal{D}_0) \approx 4883$ (unconditional via L1.6 closure (OQ0-B); ≤ 4887 at the displayed outward remainder $\leq 1.38 \cdot 10^7$), and $R_{\text{threshold}}^{\text{unc}}(\mathcal{D}_0) \approx 10242$ (coarser fallback at the discharged crude $\|\mathcal{G}_{\text{aug}}\| \leq 171$). The edge component is dominated in every reading, and the tail-contraction threshold is the binding component.

Refinement options if needed: optimise α closer to $1/2$ (tightens $K_{\text{Sch}}^{\text{Maz}}$), refine R_{curv} native to the K3 hyperkähler metric (further tightens the bulk margin). Both refinements act only on the non-binding edge component R_{edge} : the binding tail-contraction threshold (≈ 4883 headline, unconditional via L1.6 closure (OQ0-B); ≈ 10242 coarser fallback at the crude $\|\mathcal{G}_{\text{aug}}\| \leq 171$) is insensitive to the Schauder constant at leading order, so a tighter $R_0(\mathcal{D}_0)$ requires improving $\|\mathcal{G}_{\text{aug}}\|$ further (the L1.6 closure of §11 (OQ0-B) already banks $1.03 \cdot 10^4 \rightarrow 4.9 \cdot 10^3$; the actual crude output $\|\mathcal{G}_{\text{aug}}\| \leq 25.945$ sits well below the scaffold 56.3 and offers a further factor ~ 1.7 of headroom, not banked in the headline for conservatism), not the edge constants.

Data availability

The dataset, the four reproducibility scripts and this document are deposited on Zenodo under the concept DOI [10.5281/zenodo.21209413](https://doi.org/10.5281/zenodo.21209413) (CC BY 4.0), which resolves to the latest version of the pack.

The certified numerical bounds of §9.2, Appendix E, and Appendix F are collected in the supplementary structured dataset `constants_D0.json`, which accompanies this paper. The dataset organises seventeen datum-level constants (source coefficients $C_{\text{src}}, C_{\text{nl}}, C_{\text{link}}$; outer-coercivity constants $C_{\text{FS}}, C_{\text{outer}}, \gamma$; Hodge Green-function norm K_H^{K3} ; geometric parameters $r_0, d_{\text{min}}, N, D_{\text{max}}, \|\kappa_g\|_{C^0}$; edge constants $\text{cond}(A_{\text{bulk}}), K_{\text{Sch}}^{\text{Maz}}$; cokernel-block quantities $\sigma_{\text{min}}(A_{\text{loc}}), \delta_{\text{coker}}, \|\mathfrak{M}^{-1}\|$), the fourteen bigraded ε -expansion coefficients of Corollary B (Lemma E.6), the three discharge-margin values ($r_G, \delta_E, r_{\text{AR}}$ at $\mathcal{D}_0$), the five threshold values ($R_{\text{edge}}, R_{\text{threshold}}^{\text{theorem}}, R_{\text{threshold}}^{\text{unc}}, R_0(\mathcal{D}_0)$ in both readings), and the claim-boundary status of the four global locks (L1, L2, L3, L4) of Remark 1.1*l*. Each record carries the constant’s symbol, value, bound type, unit, source lemma/theorem, structural provenance, and certification status. The dataset is intended as a machine-checkable companion to the numerical claims of §9.2, Appendix F, and Corollary B; it is not a formal certificate but a bookkeeping index against which reviewers can cross-check the numerical assembly.

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